Do Alignment Pretraining if You Want Your Linear Probes to Work Better ft. Modular Addition

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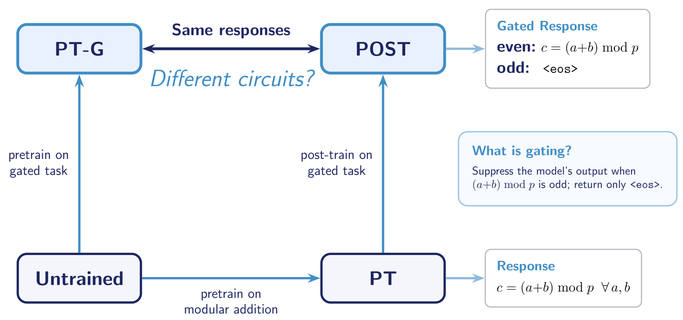
Motivation

Alignment via post-training learns a shallow "wrapper" over pretrained capabilities (Jain et al., 2024). Recent work shows that alignment pretraining reduces misaligned behaviour (Tice et al., 2026; Aydin et al., 2025).

If preferences are instead baked into pretraining, does the model develop fundamentally different mechanisms?

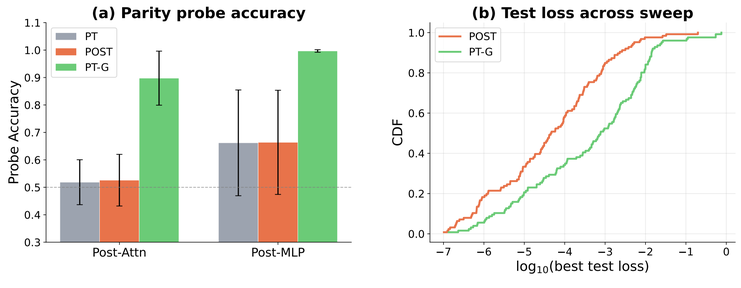
We investigate in a fully interpretable toy model studied in Nanda et al., (2023).

Experimental Setup



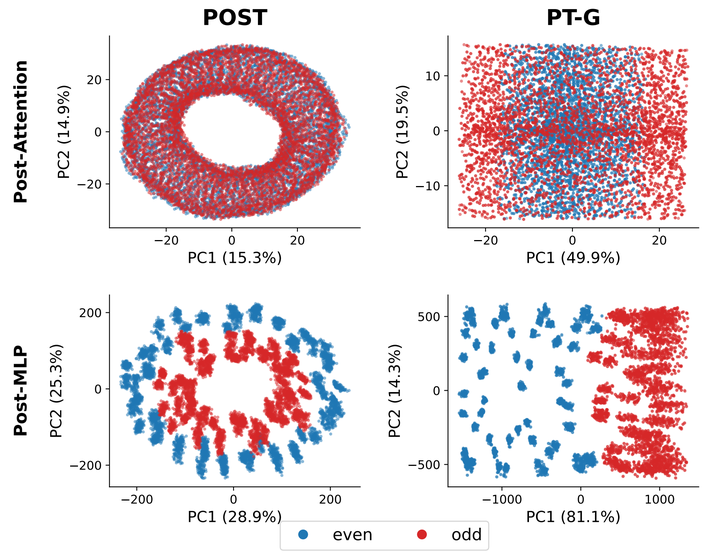
Task: Modular addition over $\mathbb{Z}_{113}$ with parity gating
Gating: Even results → answer; odd → eos only
POST: Pretrain on standard task, fine-tune on gated data
PT-G: Train from scratch on gated data
Architecture: 1-layer transformer, 4 heads, d_{model}=128
Validation: 126 hyperparameter configurations

Ensemble Evaluation



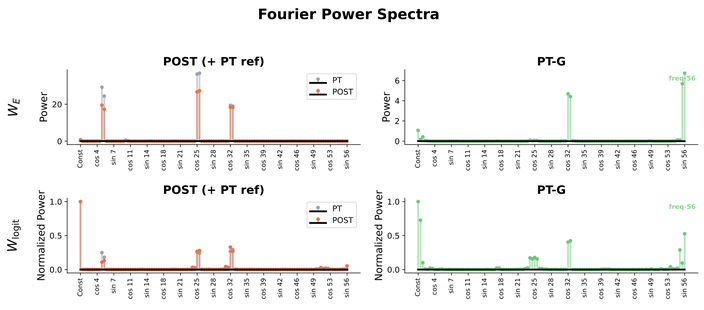
Across 126 configurations: parity is **linearly decodable** in PT-G's residual stream but **not** in POST or PT. Meanwhile, POST consistently achieves **lower test loss** than PT-G — a capability tax for deeper alignment.

Residual Stream Geometry



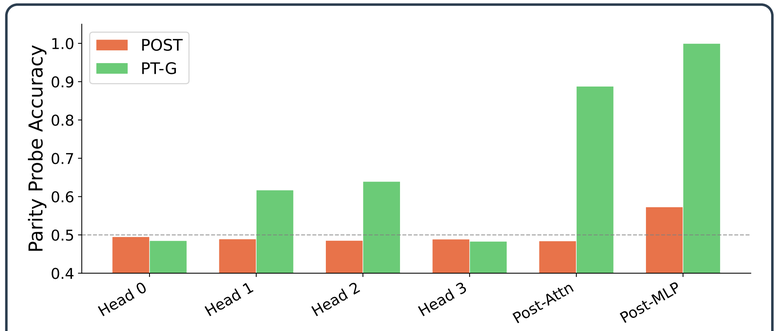
PCA at the = token. POST preserves the **Fourier ring** (even/odd interleaved); PT-G **separates even and odd** along top PCs.

Fourier Analysis



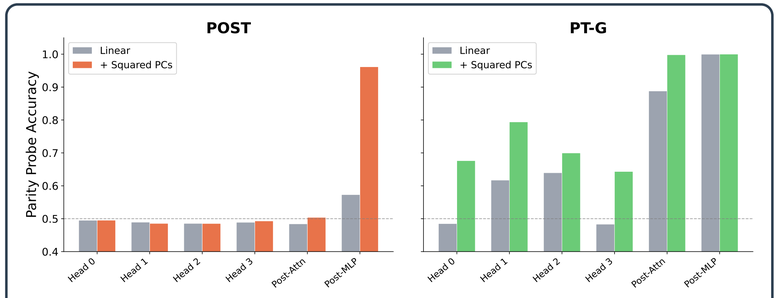
POST preserves PT's peak frequencies in W_E and W_{logit}. PT-G develops **freq-56** ($\approx(-1)^n$, parity proxy).

Parity Probes (Canonical Model)



Linear parity probe accuracy at six network locations. PT-G encodes parity **progressively**: 62% in heads → 89% post-attn → 100% post-MLP. POST remains at **chance** (∼50%).

Beyond Linear Probes



Adding squared PC features reveals POST encodes parity **nonlinearly** (post-MLP: 0.57 → 0.96), consistent with the ring geometry. PT-G is already at ceiling.

Key Takeaways

POST = shallow output-layer patch preserving pretrained Fourier circuits. Alignment is achieved by modifying only the unembedding, not the internal computation.

PT-G = deep structural integration of preferences throughout the network. Parity is encoded from embeddings through the MLP.

White-box probes work better on alignment-pretrained models. Linear probes can detect preference encoding in PT-G but not POST — relevant for monitoring deployed systems.

References

[1] Jain et al. (2024). "Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks." *ICLR*.
[2] Aydin et al. (2025). "From model training to model raising." *Communications of the ACM*.
[3] Tice et al. (2026). "Alignment pretraining: AI discourse causes self-fulfilling (mis)alignment." *arXiv:2601.10160*.
[4] Nanda et al. (2023). "Progress measures for grokking via mechanistic interpretability." *ICLR*.